

OOPS Concepts:



SIMATS Online Programming Lab
JAVA PROGRAMMING

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QUESTIONS 20 / 20 completed

Bank

Write a Java program to implement a basic banking system using inheritance. Create a parent class called Account with attributes such as account number, account holder name, and balance. Then, create child classes such as SavingsAccount and CheckingAccount that inherit from the Account class and

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class First
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int a=sc.nextInt();
8         int b=sc.nextInt();
9         System.out.println("Before Swapping");
10        System.out.println("A = " + a + " , B = " + b);
11        a=b;
12        b=a-b;
13        a=a-b;
14        System.out.println("After swapping");
15        System.out.println("A = " + a + " , B = " + b);
16    }
17 }
```

OUTPUT

After swapping
A =20 , B = 10

INPUT

10
20



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QUESTIONS 20 / 20 completed

Employee

Write a Java program to implement a basic employee management system using inheritance. Create a parent class called Employee with attributes such as name, id, and salary. Then, create child classes such as Manager and SalesPerson that inherit from the Employee class and

PROGRAM EDITOR

Java Run Save

```
1 import java.util.Scanner;
2 public class second
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int num;
8         num=sc.nextInt();
9         int fact=1;
10        for(int i=1;i<=num;i++)
11        {
12            fact*=i;
13        }
14        System.out.println("factorial = "+ fact);
15    }
16 }
```

OUTPUT

factorial = 1

INPUT

5



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QUESTIONS 20 / 20 completed

Animal-1

Write a Java program to implement a basic animal classification system using inheritance. Create a parent class called Animal with attributes such as name and habitat. Then, create child classes such as Mammal, Reptile, and Bird that inherit from the Animal class and have their own unique methods for action

PROGRAM EDITOR

Java Run Save

```
1 public class Main {
2
3     public static void main(String[] args) {
4
5         Mammal mammal = new Mammal("Lion", "Grassland");
6         Reptile reptile = new Reptile("Snake", "Forest");
7         Bird bird = new Bird("Eagle", "Mountains");
8
9         System.out.println("Mammal:");
10        mammal.display();
11        mammal.eat();
12        mammal.reproduce();
13
14        System.out.println();
15
16        System.out.println("Reptile:");
17        reptile.display();
18        reptile.eat();
19        reptile.reproduce();
20
21        System.out.println();
22    }
23 }
```

OUTPUT

Mammal:
Name: Lion
Habitat: Grassland
Mammal eats food.

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Shapes-1

Write a Java program to implement a basic shape hierarchy using inheritance. Create a parent class called Shape with attributes such as color and area. Then, create child classes such as Circle, Square, and Triangle that inherit from the Shape class and have their own unique methods for calculation area and

PROGRAM EDITOR

Java Run Save

```
1 public static void main(String[] args) {
2
3     Circle circle = new Circle("Red", 5);
4     Square square = new Square("Blue", 4);
5     Triangle triangle = new Triangle("Green", 3, 4, 5);
6
7     System.out.println("Circle:");
8     circle.display();
9     System.out.println("Area: " + circle.calculateArea());
10    System.out.println("Perimeter: " + circle.calculatePerimeter());
11
12    System.out.println();
13
14    System.out.println("Square:");
15    square.display();
16    System.out.println("Area: " + square.calculateArea());
17    System.out.println("Perimeter: " + square.calculatePerimeter());
18
19    System.out.println();
20
21    System.out.println();
22 }
23 }
```

OUTPUT

Circle:
Color: Red
Area: 78.53981633974483
Perimeter: 31.41592653589793

INPUT

Your INPUT goes here!

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Library

Write a Java program to implement a basic library management system using inheritance. Create a parent class called Item with attributes such as title and author. Then, create child classes such as Book and DVD that inherit from the Item class and have their own unique attributes and methods.

```

1 public class Main {
2     public static void main(String[] args) {
3
4         Book book = new Book("Java Programming", "James Gosling", 500);
5         DVD dvd = new DVD("Avengers", "Marvel Studios", 120);
6
7         System.out.println("Book Details:");
8         book.display();
9         book.borrowItem();
10        book.returnItem();
11
12        System.out.println();
13
14        System.out.println("DVD Details:");
15        dvd.display();
16        dvd.borrowItem();
17        dvd.returnItem();
18
19    }
20 }

```

OUTPUT

```

Book Details:
Title: Java Programming
Author: James Gosling
Pages: 500
Book borrowed successfully.

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

University

Write a Java program to implement a basic university management system using inheritance. Create a parent class called Person with attributes such as name and age. Then, create child classes such as Student and Professor that inherit from the Person class and have their own unique attributes and methods.

```

1 public class Main {
2     public static void main(String[] args) {
3
4         Student student = new Student("Tharun", 20, "Computer Science");
5         Professor professor = new Professor("Dr. Kumar", 45, "Java Programming");
6
7         System.out.println("Student Details:");
8         student.display();
9         student.manageCoursework();
10
11        System.out.println();
12
13        System.out.println("Professor Details:");
14        professor.display();
15        professor.teachClass();
16
17    }
18 }
19
20 class Person {
21
22 }

```

OUTPUT

```

Student Details:
Name: Tharun
Age: 20
Course: Computer Science
Student is managing coursework.

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Games

Write a Java program to implement a basic game system using inheritance. Create a parent class called Game with attributes such as title and genre. Then, create child classes such as ActionGame and PuzzleGame that inherit from the Game class and have their own unique attributes and methods.

```

1 public class Main {
2     public static void main(String[] args) {
3
4         ActionGame actionGame = new ActionGame("Warrior Battle", "Action", 5);
5         PuzzleGame puzzleGame = new PuzzleGame("Brain Challenge", "Puzzle", 20);
6
7         System.out.println("Action Game:");
8         actionGame.display();
9         actionGame.play();
10
11        System.out.println();
12
13        System.out.println("Puzzle Game:");
14        puzzleGame.display();
15        puzzleGame.solve();
16
17    }
18 }
19
20 class Game {
21
22 }

```

OUTPUT

```

Action Game:
Title: Warrior Battle
Genre: Action
Difficulty Level: 5
Action game is being played.

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Restaurant

Write a Java program to implement a basic restaurant management system using inheritance. Create a parent class called Menu with attributes such as name and price. Then, create child classes such as Appetizer and Entree that inherit from the Menu class and have their own unique attributes and methods.

```

1 // Mean , Median and Mode
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         int arr[]={16,17,18,16,20,23,29,22};
7         result r= new result();
8         r.mean(arr);
9         r.median(arr);
10        r.mode(arr);
11    }
12 }
13 class result
14 {
15     void mean(int arr[])
16     {
17         int sum=0;
18         for(int i=1;i<arr.length;i++)
19         {
20             sum+=arr[i];
21         }
22     }
23 }

```

OUTPUT

```

Mean:18.0
Median: 20.0
mode= 16

```

INPUT

16 16 17 18 19 20 22
 23

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QUESTIONS 20 / 20 completed

Rental

Write a Java program to implement a basic car rental system using inheritance. Create a parent class called Vehicle with attributes such as make and model. Then, create child classes such as Sedan and SUV that inherit from the Vehicle class and have their own unique attributes and methods for rental.

```

1 public class Vehicle {
2     String make;
3     String model;
4     Vehicle(String make, String model) {
5         this.make = make;
6         this.model = model;
7     }
8     void displayDetails() {
9         System.out.println("Make: " + make);
10        System.out.println("Model: " + model);
11    }
12    public static void main(String[] args) {
13
14        Sedan sedan = new Sedan("Toyota", "Camry", 4);
15        sedan.displayDetails();
16        sedan.rentSedan();
17        System.out.println();
18    }
19 }

```

OUTPUT

```

----- Sedan Details -----
Make: Toyota
Model: Camry
Sedan rented successfully.
Number of doors: 4

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Animal-2

Write a Java program to demonstrate polymorphism with abstract classes. Create an abstract class called Animal with abstract methods such as eat() and sleep(). Then, create child classes such as Dog and Cat that inherit from the Animal class and implement their own versions of the methods.

```

1 public class Main {
2     public static void main(String[] args) {
3         Animal animal1 = new Dog();
4         Animal animal2 = new Cat();
5
6         System.out.println("Dog:");
7         animal1.eat();
8         animal1.sleep();
9
10        System.out.println("Cat:");
11        animal2.eat();
12        animal2.sleep();
13    }
14 }
15
16 abstract class Animal {
17     abstract void eat();
18     abstract void sleep();
19 }

```

OUTPUT

```

Dog:
Dog eats bones
Dog sleeps in the kennel

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Overloading

Write a Java program to demonstrate method overloading with variable arguments. Create a method called sum() that takes in a variable number of integers and returns their sum. Then, create an overloaded version of the sum() method that takes in a variable number of doubles and returns their sum.

```

1 public class Sum {
2     int sum(int... numbers) {
3         int total = 0;
4
5         for (int number : numbers) {
6             total += number;
7         }
8         return total;
9     }
10    double sum(double... numbers) {
11        double total = 0;
12
13        for (double number : numbers) {
14            total += number;
15        }
16        return total;
17    }
18    public static void main(String[] args) {
19
20    }
21 }

```

OUTPUT

```

Sum of integers: 100
Sum of doubles: 61.5

```

INPUT

Your INPUT goes here!

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QUESTIONS 20 / 20 completed

Person

Write a Java program to create a class called Person with a constructor that takes in a name and age, and a method to print out the person's name and age.

```

1 public class Person {
2     String name;
3     int age;
4     Person(String name, int age) {
5         this.name = name;
6         this.age = age;
7     }
8     void printDetails() {
9         System.out.println("Name: " + name);
10        System.out.println("Age: " + age);
11    }
12
13    public static void main(String[] args) {
14        Person person = new Person("Tharun", 20);
15        person.printDetails();
16    }
17 }
18 }

```

OUTPUT

```

Name: Tharun
Age: 20

```

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Rect-Const

RECT-CONST

Write a Java program to create a class called Rectangle with a constructor that takes in the length and width of the rectangle, and a method to calculate and return the area of the rectangle.

```
1 public class Rectangle {
2     double length;
3     double width;
4
5     Rectangle(double length, double width) {
6         this.length = length;
7         this.width = width;
8     }
9     double calculateArea() {
10        return length * width;
11    }
12
13    public static void main(String[] args) {
14        Rectangle rectangle = new Rectangle(10, 5);
15        System.out.println("Length: " + rectangle.length);
16        System.out.println("Width: " + rectangle.width);
17        System.out.println("Area: " + rectangle.calculateArea());
18    }
19 }
20 }
```

OUTPUT

Length: 10.0
Width: 5.0
Area: 50.0

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Bank-Const

BANK-CONST

Write a Java program to create a class called BankAccount with a constructor that takes in an account number and an initial balance, and methods to deposit and withdraw money from the account.

```
1 public class BankAccount {
2     String accountNumber;
3     double balance;
4     BankAccount(String accountNumber, double initialBalance) {
5         this.accountNumber = accountNumber;
6         this.balance = initialBalance;
7     }
8     void deposit(double amount) {
9         balance = balance + amount;
10        System.out.println("Deposited: " + amount);
11    }
12    void withdraw(double amount) {
13        if (amount <= balance) {
14            balance = balance - amount;
15            System.out.println("Withdrawn: " + amount);
16        } else {
17            System.out.println("Insufficient balance");
18        }
19    }
20    void displayBalance() {
21        System.out.println("Account Number: " + accountNumber);
22    }
23 }
```

OUTPUT

Account Number: 123456789
Balance: 5000.0
Deposited: 2000.0
Withdrawn: 1500.0

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Car-Const

CAR-CONST

Write a Java program to create a class called Car with a constructor that takes in the make, model, and year of the car, and a method to print out the car's make, model, and year.

```
1 public class Car {
2     String make;
3     String model;
4     int year;
5     Car(String make, String model, int year) {
6         this.make = make;
7         this.model = model;
8         this.year = year;
9     }
10    void printDetails() {
11        System.out.println("Make: " + make);
12        System.out.println("Model: " + model);
13        System.out.println("Year: " + year);
14    }
15
16    public static void main(String[] args) {
17        Car car = new Car("Toyota", "Camry", 2024);
18        car.printDetails();
19    }
20 }
21 }
```

OUTPUT

Make: Toyota
Model: Camry
Year: 2024

INPUT

Your INPUT goes here!



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QUESTIONS 20 / 20 completed

Student-Const

STUDENT-CONST

Write a Java program to create a class called Student with a constructor that takes in a name, ID number, and a list of grades, and methods to calculate and return the student's average grade and letter grade.

```
1 public class Student {
2     String name;
3     int id;
4     List<Integer> grades;
5     Student(String name, int id, List<Integer> grades) {
6         this.name = name;
7         this.id = id;
8         this.grades = grades;
9     }
10
11    double getAverageGrade() {
12        int sum = 0;
13        for (int grade : grades) {
14            sum += grade;
15        }
16        return (double) sum / grades.size();
17    }
18
19    void printDetails() {
20        System.out.println("Student Name: " + name);
21        System.out.println("Student ID: " + id);
22        System.out.println("Average Grade: " + getAverageGrade());
23        System.out.println("Letter Grade: " + getLetterGrade());
24    }
25 }
```

OUTPUT

Student Name: Tharun
Student ID: 101
Average Grade: 86.6
Letter Grade: B

INPUT

Your INPUT goes here!